How much evidence is enough? A Bayesian rare-event reading of robotaxi crash data

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Data: NHTSA Standing General Order 2021-01 ADS crash reports (Jul 2021 – Sep 2026) · Waymo Safety Impact hub files: crash list, rider-only miles and benchmarks through Jun 2026

Bottom line. Rebuilt independently from NHTSA's public crash reports, Waymo's rider-only record shows injury, airbag and serious-injury crash rates 82–95% below the human benchmark. A nuclear-PRA Bayesian model reproduces Waymo's published reductions to within 0.3 points when given the same counts. The largest remaining uncertainty is not in the crash counts but in the corrections applied to the human benchmark: Waymo's own four benchmark versions give 68.5–82.1% for any-injury crashes. For a new operator, a serious-injury claim takes about **9 million miles with zero serious crashes**.

1. Independent reconstruction matches Waymo's crash list

From 3,890 public report rows (3,450 after keeping the latest version of each), an automated pipeline isolated **1,934 Waymo rider-only crashed vehicles** in the five benchmarked metros. **1,925 (99.5%)** are in Waymo's published crash list.

Outcome counts differ in ways consistent with Waymo's manual review. Waymo flags 20 more injury vehicles, 17 of them with "Unknown" severity in the SGO data, and 9 more airbag deployments. Its serious-injury set excludes 3 SGO "Serious" cases, consistent with its police-report confirmation.

Outcome	This analysis	Waymo	Only Waymo	Only ours
Any injury	166	182	20	4
Airbag, any vehicle	72	80	9	1
Serious or worse	6	3	0	3

Outcome flags against Waymo's list (SGO era, five metros).

2. Same evidence, same answer, with richer uncertainty statements

Each outcome is a Poisson process in miles. With a Jeffreys prior, the posterior rate is Gamma(k + ½, 271.3 M mi), and the reduction is $R = 1 - \lambda/\lambda_H$.

Outcome	This analysis: median (95% CrI)	Bayesian, Waymo's counts	Waymo published (95% CI)	P(reduction ≥ 80%)*
Serious injury or worse	89.3% (78.6–95.7%), k = 6	94.5% (86.1–98.5%), k = 3	94.8% (83.2–99.2%)	≥ 0.95
Airbag, any vehicle	83.5% (79.4–87.0%), k = 72	81.7% (77.4–85.4%), k = 80	81.7% (76.6–86.0%)	≥ 0.95
Any injury reported	83.8% (81.2–86.1%), k = 166	82.1% (79.4–84.6%), k = 183	82.1% (78.9–85.0%)	≥ 0.95

*This analysis' counts: the 5th percentile of the reduction is 80.6%, 80.1% and 81.6% respectively, so each outcome has a 95% probability of a reduction of at least 80%.

The gap between the solid and dashed curves is data curation; the gap between the dashed curve and the orange marker is the inference method. With identical counts the two methods agree. The Bayesian version adds direct probability statements, such as a 0.993 probability that the serious-injury reduction exceeds 75%. It also gives usable estimates where frequentist point estimates are uninformative. For Atlanta (0 serious crashes in 8.6 M mi), Waymo reports "100% reduction" with an interval reaching −87%. The posterior gives a median of 88% and P(better) = 0.95; the result is prior-sensitive (0.86 under a flat prior).

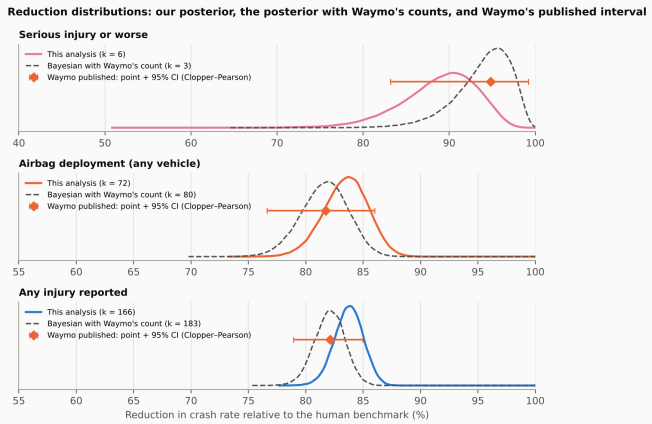


Figure 1. Reduction distributions. Solid: this analysis. Dashed: the same model with Waymo's verified counts. Orange: Waymo's published point estimate and interval.

3. Where the uncertainty really lives: the human benchmark

Waymo publishes four any-injury benchmarks: observed or corrected for under-reporting, with or without spatial re-weighting. With the same 183 crashes they give:

Benchmark version	IPMM	Reduction
Observed, unadjusted	2.14	68.5%
Observed, spatially adjusted	2.58	73.9%
Corrected, unadjusted	3.14	78.5%
Corrected, adjusted (headline)	3.77	82.1%

The versions are not equally credible: uncorrected police data undercount injuries, so Waymo's headline choice is well motivated. But the 32% under-reporting share and the spatial multipliers are themselves uncertain, and that uncertainty is not carried into the headline interval (78.9–85.0%). The spread across versions (14 points) is more than twice that interval. A bounding mixture gives 65–84%. The benefit is robust; the headline precision depends on unquantified corrections.

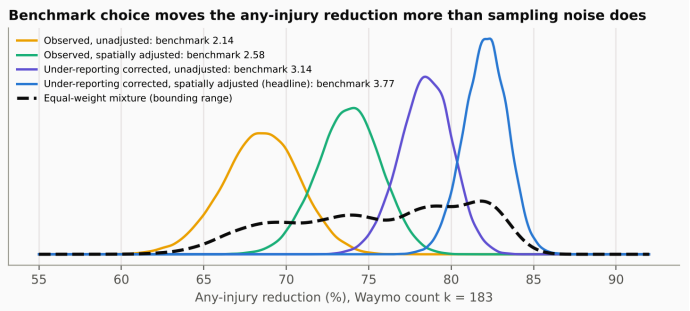


Figure 2. Any-injury reduction under each of Waymo's benchmark versions, with an equal-weight bounding mixture (black dashed).

4. How many miles does a claim need?

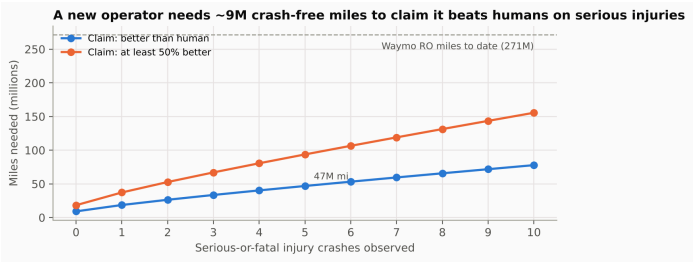


Figure 3. Miles needed for a 95%-confidence serious-injury claim, by number of serious crashes observed.

Inverting the posterior, the exposure for a 95% claim is $M^* = Q_{\text{Gamma}(k+½,1)}(0.95) / (f \cdot \lambda_H)$, where f = 1 for "better than human" and f = 0.5 for "at least 50% better".

- **Serious injury:** 9.1 M mi with zero events, 18.6 M with one, 46.8 M with five. Atlanta (8.6 M mi, 0 events) sits just below the threshold.
- **Power:** a fleet that is truly 80% safer reaches 80% power at 18.5 M mi; a fleet 50% safer needs 82.5 M mi.
- **Lower-severity outcomes** are informative early: 0.5–1.2 M mi with zero events for any-injury and airbag claims, which makes them the right leading indicators for a scaling fleet.

Zoox, Tesla, Avride and May Mobility file SGO reports but publish no comparable rider-only mileage, so their counts cannot yet support rate claims either way.

5. A PRA view of the crashes

- **Being struck dominates.** 62% of Waymo rider-only crashes happen while the vehicle is stopped or parked. With similar injury probabilities (8.1% stationary, 9.5% moving), they produce 0.36 of the 0.62 injury crashes per million miles. The dominant initiating events come from other road users.
- **Crash partner sets severity.** P(injury | crash) is 48% for cyclists, pedestrians and scooters, 20% for motorcycles, 10% for light vehicles and 2% for heavy trucks and buses.
- **Human reliability moves, it does not vanish.** In 5 of Tesla's 36 SGO narratives, a remote-assistance operator was controlling the vehicle at or just before contact. We found no explicit model of such actions in Waymo's published retrospective methodology; nuclear PRA models the analogous actions outside the main control room with dedicated HRA methods.

Limits. SGO data are self-reported and partly redacted, and the reporting rules changed in 2025. This analysis cannot reproduce Waymo's manual verification. It adopts Waymo's benchmarks and quantifies, rather than replaces, their uncertainty. Eight early-2021 crashes were likely misclassified as rider-only (8 of 1,934). Results for zero or one event are prior-sensitive. **Full report and code:** 24-page technical report with methodology, 16 figures and 21 tables; Python scripts reproduce every number. Sources: NHTSA SGO 2021-01 data; Waymo Safety Impact hub; Kusano et al. 2024, 2025; Scanlon et al. 2024; Chen et al. 2024.